

SIMATS ENGINEERING
ASSESSMENT TOOL 1 - Low (Pseudo-code Writing)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

QUESTIONS: 10

Total Marks: 100

ANNEXURE A

PSEUDO-CODE WRITING

Code of Conduct:

I ____gadiputi kotla vyshnavi____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

GK.VYSHNAVI.

CO 1 Assessment tool - 1

Pseudo code

1. Student Management System.

Start

class BOOK

private: title author, ISBN, is available.

Issue Book()

Return Book()

Display Details()

create BOOK 1, create BOOK 2, BOOK 3

Issue BOOK 1

Return BOOK 1

Display all books

Display a Stop.

2. library book management.

Start

create class BOOK

private: name, rollno, marks

Input name, rollno, marks

avg = marks

If avg ≥ 40

print pass

else

print fail

end if

stop

3. Employee Salary Calculation.

Start

create class employee

private: empId, name, basic pay

gross = basic pay + HRA + DA

net = gross - PF - Tax

Display gross and net salary
Stop.

4. Shape area calculator:

Start
Create shape class
circle $\rightarrow$ area = $3.14 \times r \times r$
Rectangle $\rightarrow$ area = $l \times b$
Triangle = area = $0.5 \times b \times h$
create objects
call calculateArea()
Display area
Stop.

5. Vehicle registration system:

Start
create vehicle class
create two wheeler and four wheeler
Input vehicle details
override display details()
Display vehicle information
Stop.

6. Bank account Hierarchy:

Start
Create Bank account
create Savings Account and current account
Input withdrawal amount
if Savings balance $\geq$ amount
with draw
else
Print Insufficient Balance
End.

currency account allows overdraft
Display balance
Stop.

7. Hospital patient Record system.

Start
create patient class
- Create in patient and outpatient
If in patient
 $\text{bill} = \text{daily charges} \times \text{days}$
Else
 $\text{bill} = \text{consultation fee}$
End If
Display bill
Stop.

8. E-commerce product catalog.

Start
create product class
create electronics, clothing, Grocery
apply category discount
 $\text{final price} = \text{price} - \text{discount}$
Display final price
~~Stop.~~

9. university staff Hierarchy

Start
create staff class
create professor and lecturer
calculate allowance
Display salary and allowance
Stop.

10. smart home device controller

```
start
create smart device
input command
if device = light
    turn on/off
else if device = thermostat
    change temperature
else
    lock/unlock door
end if
display status
stop
```